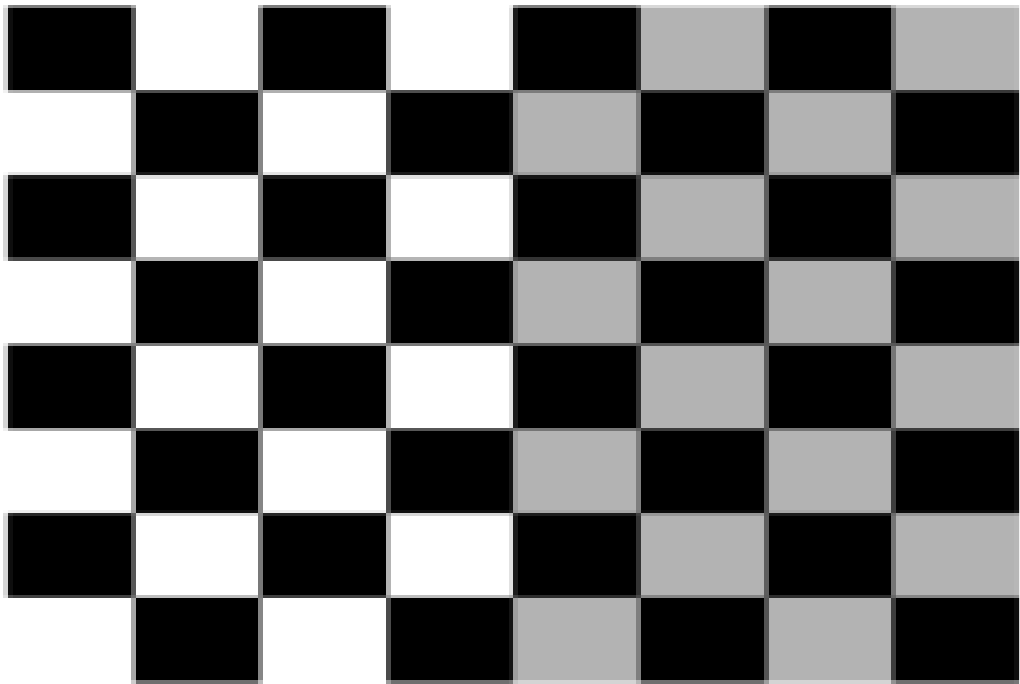


Type of effects

- There are two type of effects
 - Checkerboard effect
 - When the no. of pixels in an image is reduced by keeping the no. of gray levels in the image constant, fine checkerboard patterns are found at the edges of the image
 - False contouring effect
 - When the no. of gray-levels in the image is low, the foreground details of the image merge with the background details of the image, causing ridge like structures. This degradation phenomenon is known as false contouring



Some Mathematical Tools

- When we multiply two images, we usually carry out array multiplication

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap & bq \\ cr & ds \end{bmatrix}$$

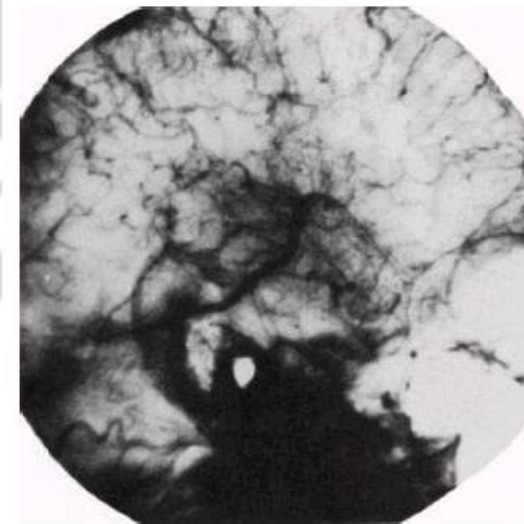
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}$$

Some Mathematical Tools (Arithmetic Operations)

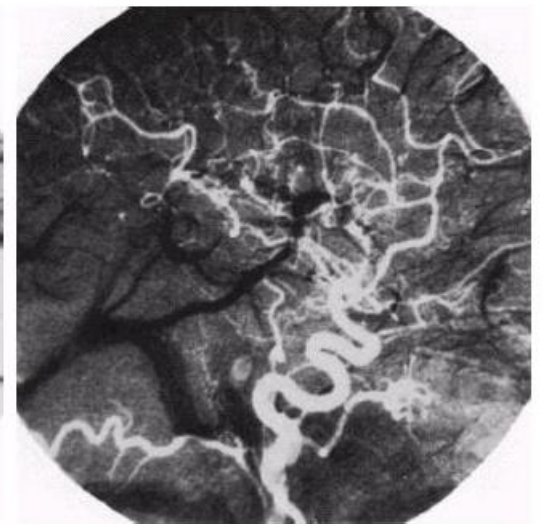
- Arithmetic operations are performed on the pixels of two or more images
- Let p and q be the pixel values at location (x, y) in first and second images respectively
 - Addition: $p + q$
 - Subtraction: $p - q$
 - Multiplication: $p \cdot q$
 - Division: p / q

Some Mathematical Tools

- Eg:
 - Blood stream is injected with a dye and X-ray images are taken before and after the injection
 - $f(x, y)$: image after injecting a dye
 - $h(x, y)$: image before injecting the dye



before injection



after injection

Some Mathematical Tools (Image Averaging)

- A noisy image:

$$g(x, y) = f(x, y) + n(x, y)$$

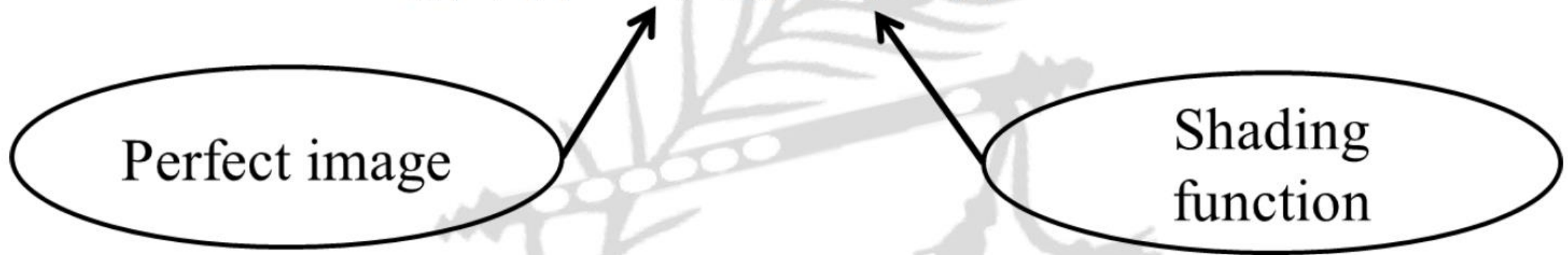
- Averaging M different noisy images:

$$\bar{g}(x, y) = \frac{1}{M} \sum_{i=1}^M g_i(x, y)$$

- As M increases, the variability of the pixel values at each location decreases
- This means that $g(x, y)$ approaches $f(x, y)$ as the number of noisy images used in the averaging process increases

Some Mathematical Tools (Shading Correction)

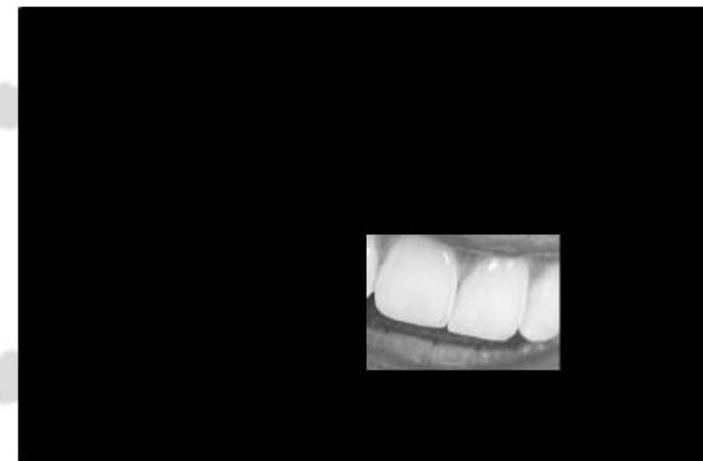
- An important applications of image division
- An imaging sensor produces image $g(x, y)$, where
- $$g(x, y) = f(x, y) h(x, y)$$



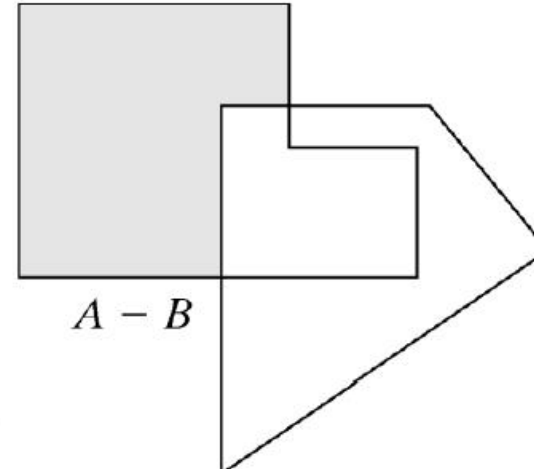
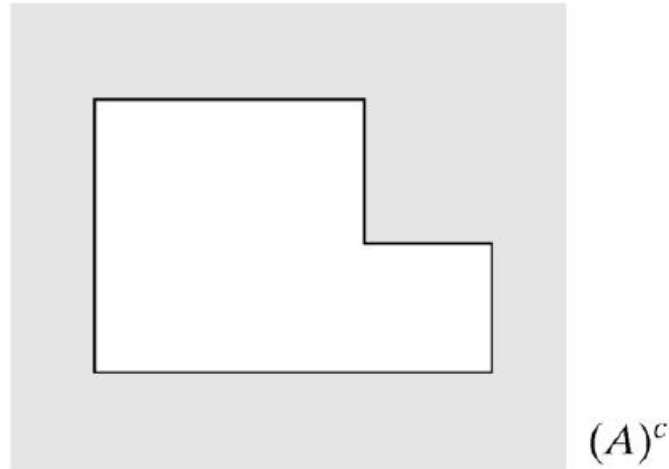
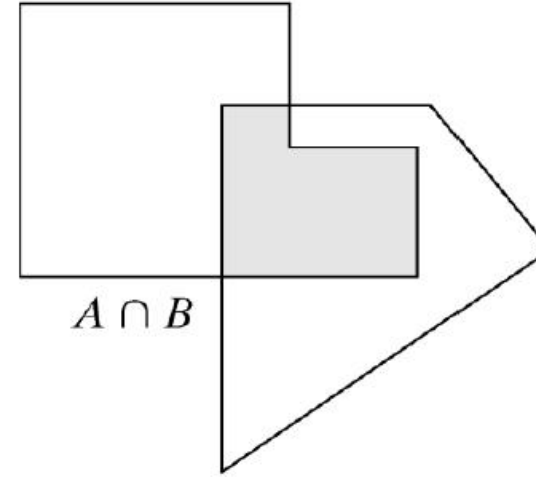
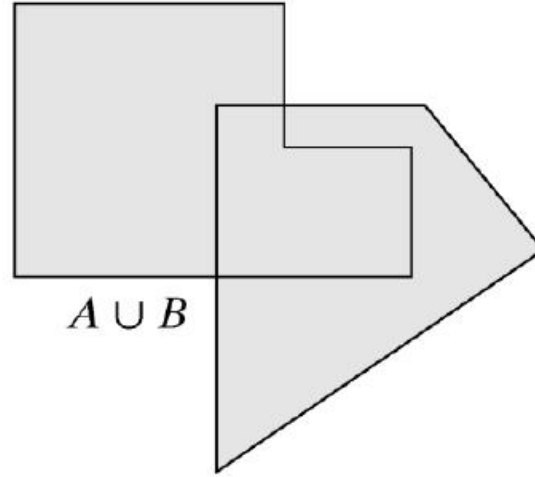
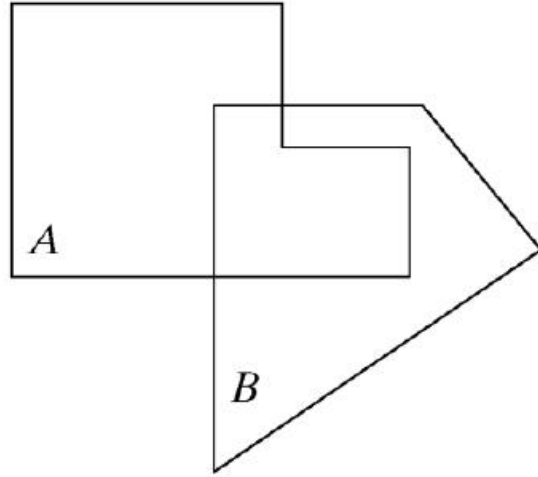
- By dividing $g(x, y)$ by $h(x, y)$, we can get the perfect image

Some Mathematical Tools (Region of Interest)

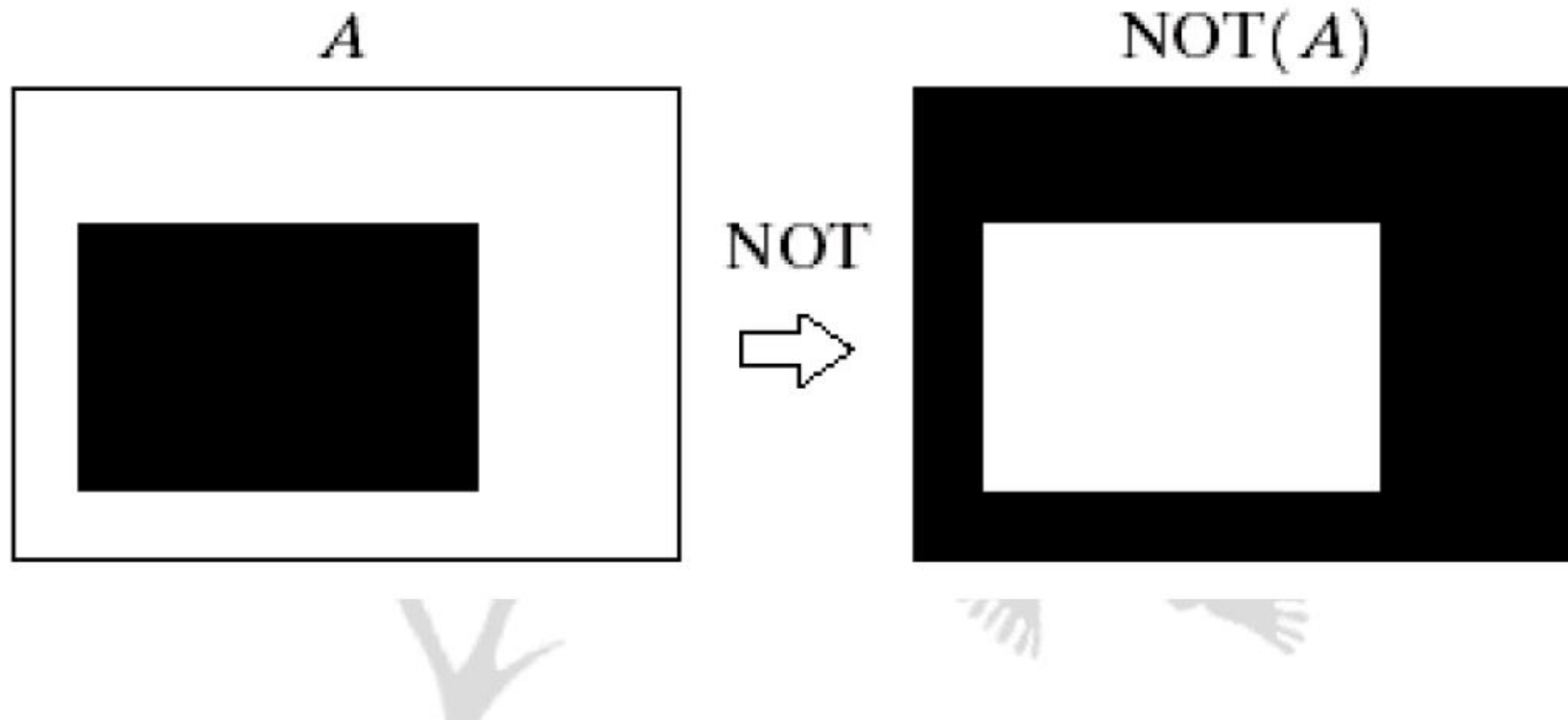
- An important applications of image multiplication



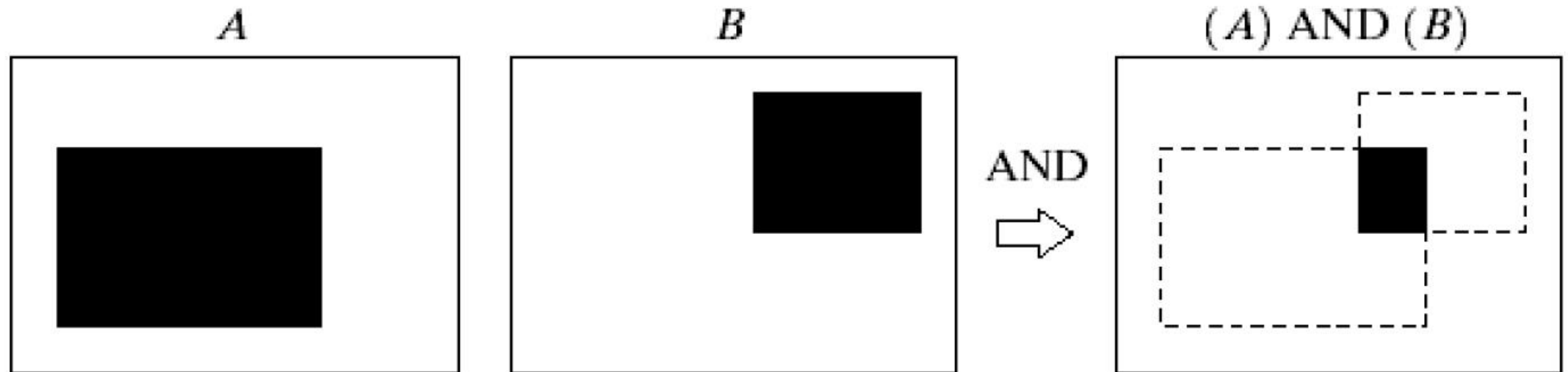
Some Mathematical Tools (Set Operations)



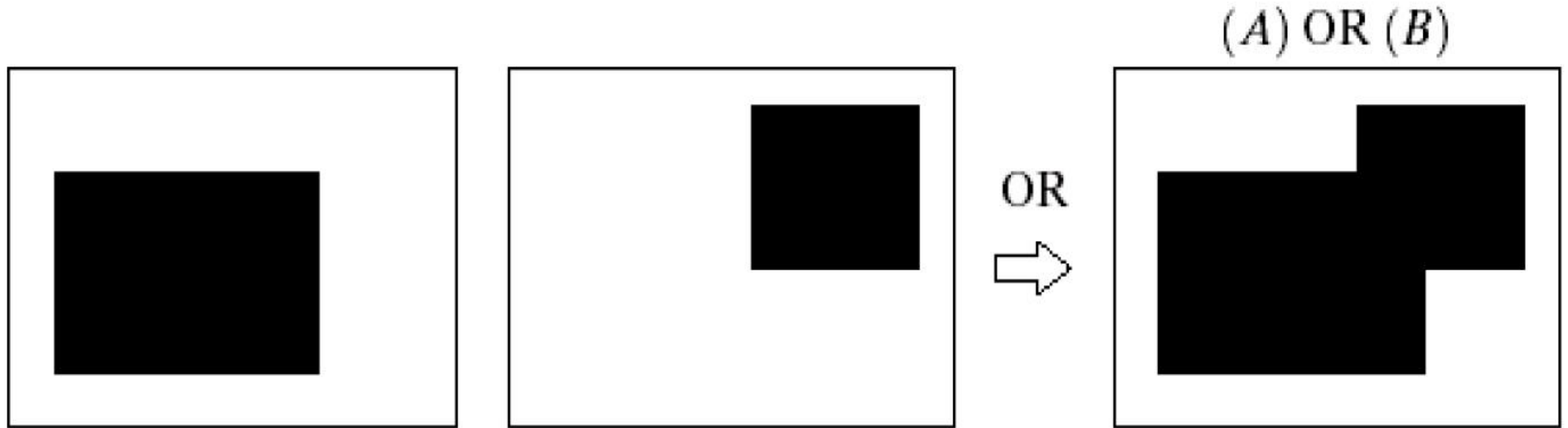
Some Mathematical Tools (Logical Operations)



Some Mathematical Tools (Logical Operations)



Some Mathematical Tools (Logical Operations)

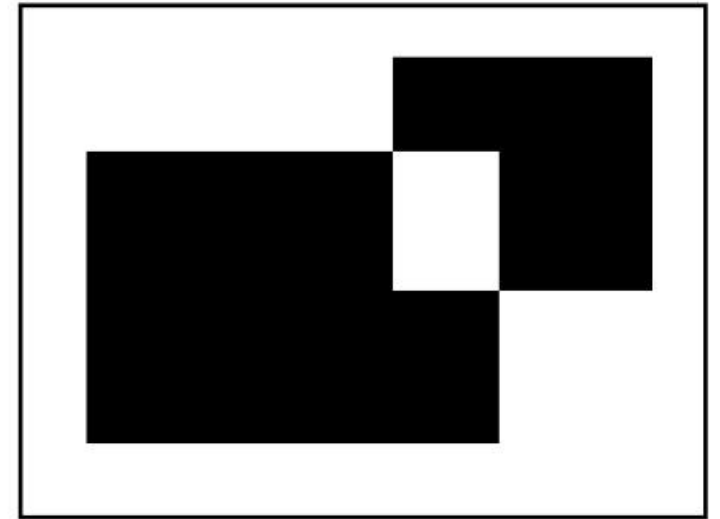


Some Mathematical Tools (Logical Operations)

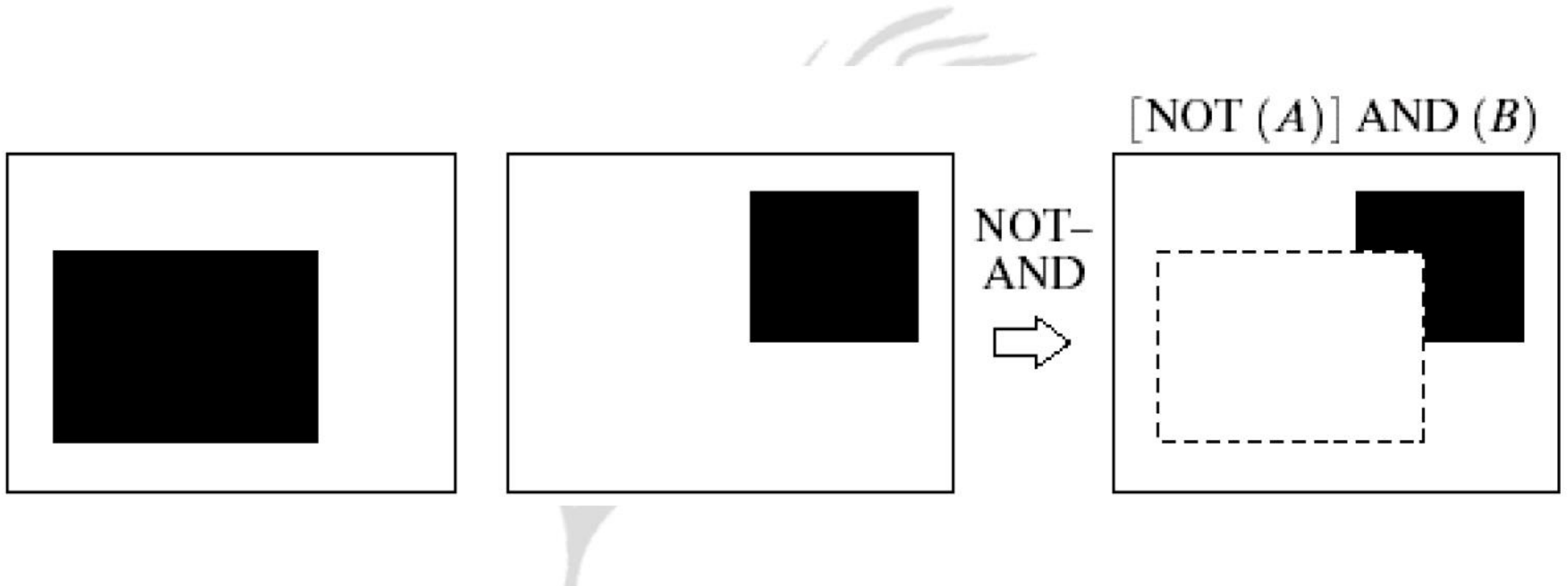


XOR
→

$(A) \text{ XOR } (B)$

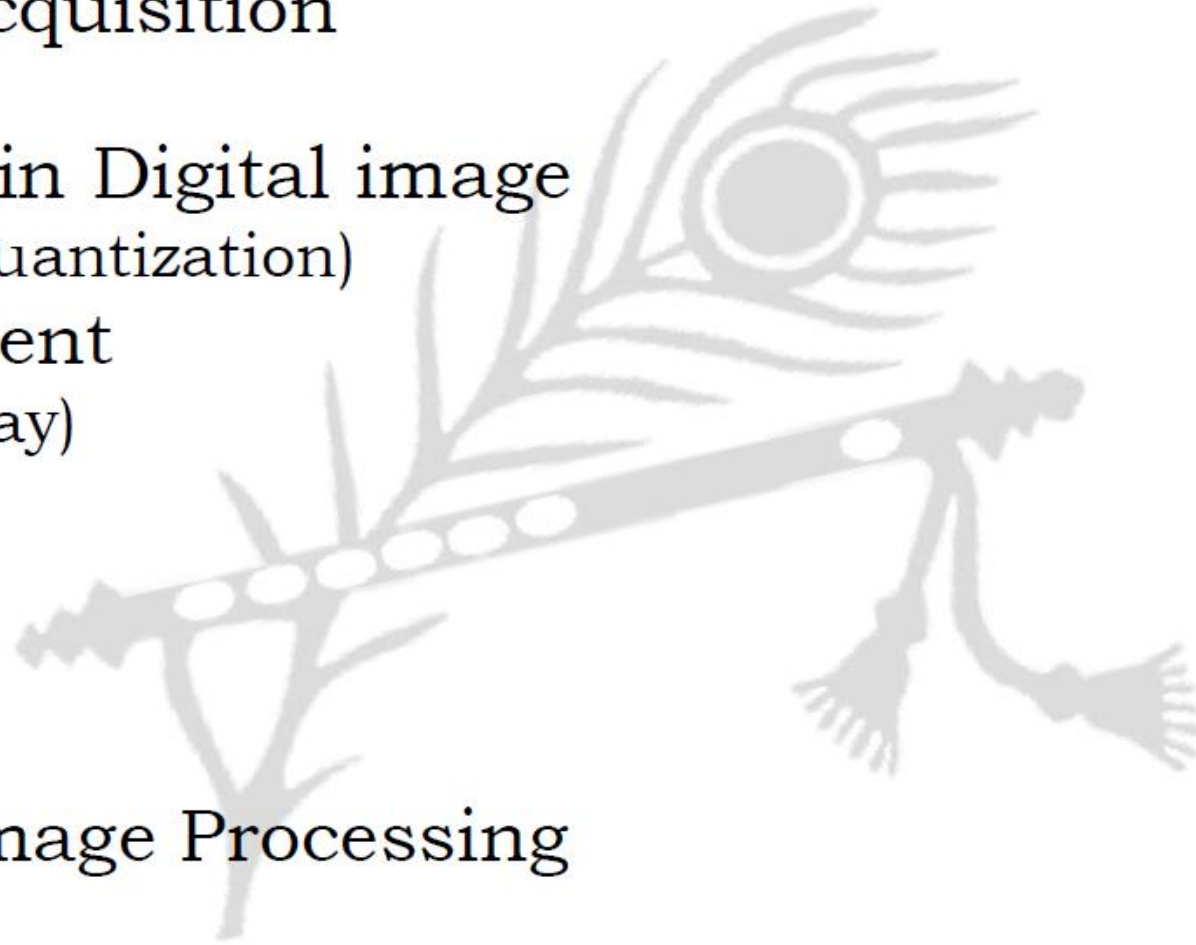


Some Mathematical Tools (Logical Operations)



Recap

- Digital Image
- Steps of Image Acquisition
 - (4 steps)
- Process involved in Digital image
 - (Sampling and Quantization)
- Sensor arrangement
 - (Single, Line, Array)
- Levels in image
 - ($L = 2^k$)
- Image size
 - ($m \times n \times k$)
- Applications of Image Processing
- Image sensing
 - (EMS)



Recap

- Types of resolutions
 - (Spatial, Spectral, Temporal, Radiometric)
- Types of effect
 - (Checkerboard, False contouring)
- Key stages of Image Processing
- Aspect ratio
 - ($r = w/h$)
- Baud rate
- Relationship between pixels
 - (Neighbour: N_4 , N_D , N_8 , Adjacency: 4-, 8-, m-, Connectivity, Path: 4-, 8-, m-, Region)
- Distance measure
 - (Euclidean, City block, Chess board)
- Set operations

Outline

- Introduction and Fundamentals:
 - Motivation and Perspective, Applications, Components of Image Processing System,
 - Element of Visual Perception, A Simple Image Model,
 - Sampling and Quantization, Some Basic Relationships between Pixels
- Intensity Transformations and Spatial Filtering:
 - Introduction, Some Basic Intensity Transformation Functions,
 - Histogram Processing, Histogram Equalization, Histogram Specification,
 - Local Enhancement, Enhancement using Arithmetic/Logic Operations
 - Basics of Spatial Filtering, Smoothing, Sharpening
- Filtering in the Frequency Domain:
 - Fourier Transform and the Frequency Domain,
 - Basis of Filtering in Frequency Domain